

Finite Element - Element – Centered - Finite Volume (FECFVM) Simulator for Incompressible 2-Phase Flow including Viscous Capillary and Gravitational Forces

Documentation of CSMP++ -based 2D simulator developed by Shaho Bazr-Afkan (2012).

Summary. The FECVM simulator is an IMPES (sequential implicit solution of wetting-phase pressure equation followed by explicit solution of the saturation transport equation), incompressible 2-phase flow solver for two-dimensional unstructured grids with lower dimensional (line-type) representations of large aspect ratio features like fractures, faults and wells. Incompressible means that the assumption is made that fluid pressure is always in a steady state. Furthermore, the simulator works with constant, pre-assigned fluid densities and viscosities.

The hybrid finite-element – finite volume numerical solution method used in the FECFVM simulator is documented in detail in the PhD thesis of Shaho Bazr-Afkan as defended at the Montanuniversitaet Leoben in 2012. It is implemented on the basis of the CSMP application programmer interface (Matthai et al. 2001, ETHZ, Tech. Reports) and summarized in Bazr-Afkan and Matthai (2012, IAMG, Salzburg). There is also a detailed article of Bazr-Afkan et al. in the ECMOR 2014, earthdoc conference volume that is going to be published in 2015 in the journal *Computational Geoscience*.

The hybrid FE-FVM discretization employed by the simulator is an extension of the FEFVM method of Geiger et al. (2004, Geofluids) and Matthai et al. (2007, Geol.Soc. London, Spec. Publ.). The important difference to these earlier schemes is that saturation is discretized as a piecewise constant element variable as are all the other material properties. Discretizing saturation on the elements is possible because conservative inter-element fluxes are achieved by solving an extra set of equations.

Per default, the simulator works with triangular element meshes output from ANSYS using its CSP interface.

Among other options, the simulator can incorporate experimentally determined capillary pressure and relative permeability curves that are supplied via a “rock type.txt” file, associating model regions identified via integer codes with RRT/SCAL data.

Solver configuration parameters are specified in the file “numerical.txt.”

The simulator can be run with different physics enabled: pressure-gradient driven flow only (viscous); viscous + gravitational (buoyant oil or gas), and or capillary spreading and counter-current imbibition across material interfaces. In this case wettability becomes key.

Runs can be interrupted, terminated and or restarted any time using the escape sequence control-^C from the command line. This signal leads to a menu of intuitive options.

Configuration of the FECFV simulator

The FECVM simulator is configured through a text-based (ASCII) configuration file ("model name"-configuration.txt and the two auxiliary files "rock_type.txt" and "numerical.txt". Using these files you have to assign material properties and essential conditions (Dirichlet) and initial (at the first time step) conditions to the model. Without such settings, the simulator will not run or produce unspecified useless results. The following describes the variables that need to be set to achieve this configuration. Note that there are a wide range of extra settings that control which variables and how often these are output. These are important because – without output – one cannot analyse the runs.

"fluid pressure" (Pa) specifies the initial pressure in the model. Since the simulator computes steady-state (instantaneously equilibrated) fluid pressure fields at each timestep, the initial pressure does not affect the results and it is better to initialize it to zero everywhere. **Attention:** A Dirichlet pressure essential condition (fixed value) needs to be applied at least to one node or model boundary. Otherwise, there is no unique solution to the integral solution for the pressure field.

Attention: The variable *"fluid pressure"* refers to the absolute pressure, which increases with depth. Thus, when a constant fluid pressure value is applied as a Dirichlet condition to a vertical feature, this leads to a distorted flow field: The pressure difference between the feature and its surroundings will be greater at its bottom than that at its top and there will be a correspondingly larger fluid flux. At the model side boundaries, this can be compensated for by applying a linear fluid pressure gradient, i.e. a higher fluid pressure value at the bottom than at the top as calculated in accordance with the fluid statics.

Attention: The applied boundary conditions may be conflicting. A producer well that intersects a constant pressure (Dirichlet) boundary is an example of this and the simulator will not run or produce meaningless results in this case. As a rule of thumb, different fixed 'fluid pressure' Dirichlet nodes should always be separated by 5 or more elements from one-another, for the simulator to show realistic behaviour. Elements with fluid volume sources and sink terms must never have simultaneous Dirichlet constraints on their nodes as these would cancel out the effects of the sources.

"saturation" (as fraction with values from 0..1) is used to specify oil saturation at those (triangular) element faces that are located on the model boundary. **Attention:** On all inflow boundaries (including injector wells) saturation must be specified using the *"saturation"* variable. Otherwise the simulator has no definition of the composition of the fluids that are coming into the model. The simulator uses the specified saturation values from the inflow boundaries to compute fluid influxes. Thus, assigning a value of zero to this variable means that only water enters through this boundary. A value of one implies 100% oil saturation. Values between zero and one also may be used. **Attention:** It is better to initialize *"saturation"* also on no-flow boundaries even if it will not be used inside the domain. This applies because inflow directions may vary with saturation over the course of a run. Note also, that boundaries with Dirichlet *"fluid pressure"* condition may act as inflow boundaries. However, outflow boundaries that shall be monitored, must not have a fixed value *"saturation"* as this would make the results meaningless.

“boundary source” ($m^3 \text{ m}^{-2} \text{ s}^{-1}$) quantifies the flux magnitude across Neumann boundaries. This assigns in- or out fluxes to the external boundaries of the model. Note, however, that a *“boundary source”* value will only have an effect where *“fluid pressure”* has not been set to a specific Dirichlet value. Good candidates are natural *“fluid pressure”* boundaries (=no-flow). **Attention:** Remember to initialize the variable *“saturation”* at boundaries, which become inflow boundaries by setting the value of *“boundary source”* to a value greater than zero. In this case, the simulator will calculate the influx for each phase by multiplying the values of *“boundary source”* with *“saturation”*. This setting will also increase *“fluid pressure”*, producing a uniform fluid injection through the boundary. **Attention:** the total injected fluid rate can be calculated by multiplying the specific *“boundary source”* value with the boundary area (if a surface) / length (in the case of a series of line elements). A negative value of *“boundary source”* creates a fluid sink and can be used to simulate an outflow boundary.

Attention: Fluid injection into- and production from within the interior of the model such as from injector and producer wells can be simulated using the following predefined source variables. Note that you must set the parameter DIVERGENCE_CORRECTION to 1, if such internal sources or sinks are applied:

“oil volume source” ($m^3 \text{ m}^{-2} \text{ s}^{-1}$) is used to simulate oil injection to elements inside the model domain. It is an element variable and determines the injected oil volume per volume of rock per second ($m^3/(m^3.s)$). Multiplying this variable by the volume of the region that this variable is assigned to, gives the total oil injection volume from that model region per second. **Attention:** This variable should be either zero or a positive real number. How to treat fluid sinks is specified further below.

“water volume source” ($m^3 \text{ m}^{-2} \text{ s}^{-1}$) is used to simulate water injection into elements in the interior of the model domain. This is also an element variable and determines the injected water volume per volume of rock per second ($m^3/(m^3.s)$). Multiplying this variable by the volume of the region that this variable has been assigned to gives the total water injection volume per second. Like the *“oil volume source”*, this variable should either be zero or a positive real number. How to model water withdrawal is subject of the next point.

“fluid volume source” ($m^3 \text{ m}^{-2} \text{ s}^{-1}$) is used to simulate fluid sinks inside the domain. Note that for sinks, the relative proportions of the fluids withdrawn are not known at the beginning of the simulation and they will also change over time. For this reason the neutral term *“fluid volume source”* is used to model such sinks. Again this is a specific quantify discretized as constant over the element quantifying the produced fluid volume per volume of rock per second ($m^3/(m^3.s)$). Multiplying the *“fluid volume source”* by the volume of the region where it is assigned gives the total produced fluid volume per second. **Attention:** This variable should be either zero or a negative real number.

Attention: Since this is an incompressible flow simulator, total inflow should balance with the total outflow, else ‘fluid pressure’ will spiral out of bounds and the *“saturation”* will begin to drift. Use a source term for the injector and constant pressure for the producer to get around this problem. This setup is also more akin to how many fields are operated.

“volume modifier” variable is used to specify the thickness (attribute) of lower dimensional elements (for instance the cross-sectional area of wells). This variable must be initialized to 1 for surface elements. When line elements are used to model fractures the *“volume”* modifier is used to apply the fracture aperture. Note that this value must be consistent with the chosen permeability.

Material Properties

“permeability” (m²) is a scalar element variable used in Darcy’s law. It must be assigned and cannot have a value of zero.

“porosity” (X) fraction, as a scalar element variable ranging from 0.00001 to 1. It must be assigned.

Fluid Properties

“density oil” (kg/m³) scalar node variable.

“density water” (kg/m³) scalar node variable.

“viscosity oil” (kg/m³) scalar node variable.

“viscosity water” (Pa.s) is a scalar node variable.

“saturation oil” (X) is a scalar element variable that requires an initial value. Make sure that the assigned value does not conflict with the *“residual saturation wetting phase”*, i.e. $1 - s_o = s_{wc}$.

“saturation water” (X) is a scalar element variable that requires an initial value. Note that anywhere in the 2-phase flow model, the values assigned to *“saturation oil”* and *“saturation water”* must add up to one.

“brooks corey parameter” (dimensionless). If the Brooks-Corey model is used (as chosen in *“Numerical.txt”*), this lambda parameter permits to account for the pore-size variation in the model. If there is no variation, the capillary pressure will be essentially saturation independent. Else $p_c(s_w)$ will increase with non-wetting phase saturation as determined by lambda. The higher its value, the flatter the $p_c(s_w)$ curve will be. Note however, that large lambda values simultaneously give rise to unrealistically distorted relative permeability curves.

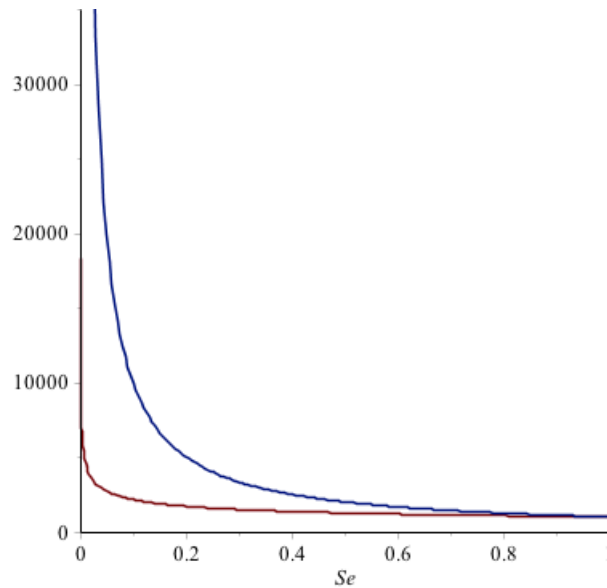


Fig. 1: Dependence of $p_c(s_w)$ on the λ value in the Brooks-Corey (1964) model. Blue curve $\lambda=1$, red curve $\lambda=3$. The blue curve is more suitable for a porous medium with large pore size variations. S_e is the effective wetting-phase saturation.

Note that the Brooks-Corey model was developed to describe drainage processes. Thus, there is an entry pressure, p_d . Instead, use imbibition $p_c(s_w)$ curves when simulating water flooding. Note also that the behaviour of the Brooks-Corey model becomes undefined for negative or zero λ values.

“*entry pressure*” (Pa) is the pressure that the non-wetting phase has to exceed in order to drain into a wetting-phase saturated porous medium. For example, to press oil into a water saturation medium grained sandstone, the oil pressure has to exceed ~ 2 kPa.

“*residual saturation wetting phase*” (X). This fraction can be used to model rock-type dependent connate water saturation, sw_c .

“*residual saturation non-wetting phase*” (X). Use this fraction to simulate rock-type dependent residual oil saturations, so_r .

“*rock type*” is used for *EXPERIMENTAL* rock fluid functions defined in the “*rock type.txt*” file. It is an integer number ranging from 0..n-1 and is used to associate RRTs with saturation functions.

Together with solver-internal variables all these enlisted parameters are specified including their physically possible ranges in the file “*IMPES-variables.txt*”. This file must be present in the same directory as the simulator at the onset of a run. **Attention:** Never modify this file.

The content of the file specifies where on the grid the variables are placed, whether they are scalars or vectors and what their SI units are. All of the variables listed can be output to VTK for visualisation if so specified in the file “*Numerical.txt*”:

Table 1: *IMPES-variables.txt*. If you are not sure that you have the correct version of this file compared it with this one.

name	MPParameter	unit	index	min.	max.	place			
fluid pressure			PF		Pa	1	1.0E+05	5.0E+08	node
permeability			K		m2	1	1.0E-21	1.0E+02	element
oil volume source			QOV		m3 m-3 s-1	1	0.	1.0E+03	element
fluid volume source			QFV		m3 m-3 s-1	1	-1.0E+03	0.	element
water volume source			QWV		m3 m-3 s-1	1	0.	1.0E+03	element
nodal fluid volume source			NQV		m3 s-1	1	-1.0E+02	1.0E+02	node
nodal oil volume source			NOV		m3 s-1	1	-1.0E+02	1.0E+02	node
nodal water volume source			NWV		m3 s-1	1	-1.0E+02	1.0E+02	node
total mobility			TM		m2	1	1.0E-19	1.0E+02	element
density oil			RHOO		kg m-3	1	1.0E-01	2.0E+03	element
density water			RHOW		kg m-3	1	1.0E-01	2.0E+03	element
viscosity oil			MUO		Pa s	1	1.0E-06	1.00E-01	element
viscosity water			MUW		Pa s	1	1.0E-06	1.60E-01	element
saturation oil			SO		X	1	-5.00E-02	1.05E+00	element
saturation water			SW		X	1	-5.00E-02	1.05E+00	element
brooks corey parameter			BC		NA	1	0.0	5.0	element
entry pressure			PD		Pa	1	1.0E+00	5.0E+07	element
residual saturation wetting phase			SWR		X	1	0.00E+00	5.00E-01	element
residual saturation non-wetting phase			SNR		X	1	0.00E+00	5.00E-01	element
gravity term			GT		kg Pa-1 s-3	2	-1.0E+05	1.0E+05	element
capillary term			CT		m s-1	2	-1.0E+05	1.0E+05	element
porosity			PHI		X	1	1.0E-05	1.00E+00	element
velocity			VT		m s-1	2	-1.0E+02	1.00E+02	element
velocity oil			VO		m s-1	2	-1.0E+02	1.00E+02	element
saturation			SO		X	1	-5.00E-02	1.05E+00	node
boundary source			NS		m3 m-2 s-1	1	-1.E+03	1.0E+03	node
divergence			DG		X	1	-1.0E+03	1.00E+03	element
volume modifier			VM		X	1	1.0E-6	1.00E+02	element
rock type			RT		X	1	0.	1.00E+02	element

Solver Configuration File “Numerical”

This primary configuration file sets runtime parameters and keywords for the FECVFM simulator. The user can also specify which variables are output to file. Note that how often variables are output to file is specified at the bottom of the model “-configuration.txt” file which is not described here because it forms part of the wider documentation of CSMP++ simulation tools.

The Numerical file is formatted and structured as follows:

- File starts with a documentation block that also briefly explains the keywords
- Comments are preceded hash(#) symbols / tags.
- The file columns are tab delimited. This means that between keywords and values should be a tab (and no additional whitespace). The order of the keywords is not important.
- For variables that are not specified explicitly by the user a default value will be used based on the other settings.
- In editing the file, it is important to press enter at the end of each line, even the last line of the file.

Description of the keywords:

RELPERM_MODEL. Possible parameter values are LINEARMODEL, BROOKSCOREY, VANGENUCHTEN or EXPERIMENTAL model. In the latter case experimentally determined relative permeability and capillary pressure curves are read into the simulator from the file “rock type” which is documented in detail further below. In this case integer codes assigned to the model regions in the configuration file are used to associate these regions with the saturation functions given in “rock type”. The default for the RELPERM_MODEL is BROOKSCOREY.

DIVERGENCE_CORRECTION could be 0 (false) or 1 (true). Default is 0. It is a solver-internal parameter that comes into play in gravity-dominated displacement processes. Attention: the divergence correction must be enabled if fluid volume sources or sinks are applied to regions in the interior of the model domain. For example, injector or producer wells.

FORMULATION could be GLOBAL or PHASE. Default is GLOBAL. This determines whether capillary pressure is included into the governing pressure equation. Refer to Aziz & Settari's (1979) book on reservoir simulation to understand the differences between these approaches.

NORM_SAT_CHANGE specifies the desired magnitude of saturation changes in the finite-element cells that are targeted in each transport step. The default value is different for different settings. If the value is increased, the simulation speed will increase, but the explicit solution of the saturation transport equation may become unstable.

MAX_SAT_CHANGE specifies the maximum saturation changes allowed in each transport step. It puts a cap on NORM_SAT_CHANGE. Default value is different for different settings.

CFL_MULTIPLIER specifies the Courant-Fredrich-Levy multiplier for transport steps. It gives an absolute global upper bound on the size of the transport steps in the model. CFL must be ≤ 1 for the explicit transport scheme used in this simulator. Default is different for different settings.

MAX_TRANSPORT_STEPS specifies the maximum number of transport steps executed before the pressure field is updated again. Default is different for different settings. Thus, when gravity-driven flow dominates over pressure-driven flow and there are many vertical highly permeable fractures, a prudent setting is warranted since the pressure field is invalidated immediately when saturation changes in the fractures. On the other hand, in a pressured-driven flow dominated situation where saturation changes have little impact on pressure, the pressure-update frequency can be much lower.

MAX_TRANSPORT_TIME_SIZE puts a limit on the model-time allowed to pass before the pressure field is updated again. Again the default depends on the flow scenario that is modeled.

SOLUTION specifies the solution method. Values can be IMPES, QEIMPES and QFIMPES. Default is improved IMPES called QEIMPES, which is an IMPES strategy with sub-stepping of the transport equation. QFIMPES applies the "active element concept" to obtain a further speedup by solving the transport equation only in those cells of the model which undergo saturation changes at the particular timestep. See Bazr-Afkan's (MUL, 2012) PhD thesis for further details. IMPES = the classical time stepping approach is the most robust one.

ACTIVATION_CRITERIA specifies criteria that control the number of active cells when the solution approach is QFIMPES. It is used to deactivate cells after the saturation front has passed through them. Default is 1.e-10.

DEACTIVATE specifies the number of transport steps that should be done before scanning the model for cells that can be deactivated. Default value is 0, meaning never to deactivate any cells.

LIMITER specifies criteria to (re)activate cells. Default is 1.e-3.

SAMG_RECONSTRUCTION specifies how often the hierarchy of grids that is established in the algebraic multigrid solver for systems (SAMG, from SCAI Fraunhofer Gesellschaft, Bonn, St. Augustin, Germany) gets updated (as opposed to reused). CSMP++ uses the SAMG solver for any of its larger linear algebraic calculations. SAMG grid-hierarchy updates become necessary when the nature of the flow patterns changes, for instance, when the saturation front spreads far away from an injector well.

keywords that control the output of variables to files readable by the Visualisation Toolkit (VTK, [kitware.com](http://www.kitware.com)). These files (extension “*.vtu” for XML formatted VTK files) can be read using the graphical interface tools *Paraview* or *MayaVi*.

OUTPUT_ONCE specifies any number of variables to the simulator, which shall be output only once at the beginning of a simulation. By default, the variable “permeability” is always output once.

OUTPUT_ALWAYS specifies those variables / properties that will be output after each pressure update. Defaults are ‘fluid pressure’, ‘saturation oil’ and ‘velocity’ which is a vector variable placed on the element.

Table 2: Example of user- input section of Numerical.txt with comments:

RELPERM_MODEL	EXPERIMENTAL
DIVERGENCE_CORRECTION	1
FORMULATION	GLOBAL
NORM_SAT_CHANGE	0.01
MAX_SAT_CHANGE	0.025
CFL_MULTIPLIER	0.95
MAX_TRANSPORT_STEPS	200
MAX_TRANSPORT_TIME_SIZE 1e3	
SOLUTION	QEIMPES
ACTIVATION_CRITERIA	1.0E-12
LIMITER	0.01
DEACTIVATION	0
SAMG_RECONSTRUCTION	100
OUTPUT_ONCE	
permeability	
fluid pressure	
velocity	
OUTPUT_ALWAYS	
saturation oil	

Known Bugs

While the simulator offers the option to use the *VANGENUCHTEN* relative-permeability model, the required variables “*van Genuchten parameter*” (scalar element property) must be added manually to the “*IMPES-variables.txt*” file before this option will run correctly.

The default and/or user-defined parameters in file “Numerical.txt” cannot possibly account for all possible simulation scenarios. Thus, saturations may start to drift if there are problems with the boundary conditions or some of the settings are too much tuned for speed. In this case, careful parameter tuning becomes necessary. However, in most cases when such issues arise, the model is

ill configured. This means that the user has to carefully check their input together with the simulator output for whether the intended simulation scenario is realistic.

File “rock type.txt”

To use data from this input file, the RELPERM_MODEL keyword in the text file “Numerical.txt” must be set to EXPERIMENTAL. To be read, the rock type file must be named 'rock type.txt' and it must be placed into the same directory as all the other files used by the simulator.

The content of “rock type.txt” are specifications of relative permeability and capillary pressure curves for the different material domains that shall be discerned in the model. As many reservoir rock types (RRTs) as necessary can be defined and there can be as many entries per rock type as is necessary to capture their saturation functions. The order of the RRT specifications in the file defines the RRT number (counting from 0 to n-1). This number must be assigned to the scalar element variable “rock type” in the configuration file of the model, model region by model region.

The number of RRT specifications is defined as the first integer value of the file. Then, for each RRT, the relative permeability values for the wetting and non-wetting phase are tabulated as a function of the absolute water saturation (the saturation endpoints (swc and 1-sor) are inferred from these tabulations). Each data block is preceded by the number of rows-1 (lines of text in the file minus 1) it contains. There must be no blank lines between them. The extra line is used to specify the derivatives of the curves at the saturation endpoints. In detail, the definition of each block is as follows:

Table 3: Input file structure for rock types

```

2 ->number of tables
21 ->number of entries for table 0 (first table) excluding the first row
-1.9  -0.4  0.1  3.7  -10557.  -519. ->derivatives at start/end for kro, krw and pc
0      1      0      5000 -> data columns: sw  kro krw pc)
0.1    0.8019 0.0001 3162.27766
0.2    0.6144 0.0016 2236.067977
0.3    0.4459 0.0081 1825.741858
0.4    0.3024 0.0256 1581.13883
0.5    0.1875 0.0625 1414.213562
0.6    0.1024 0.1296 1290.994449
0.7    0.0459 0.2401 1195.228609
0.8    0.0144 0.4096 1118.033989
0.9    0.0019 0.6561 1054.092553
1      0      1      1000
6 ->number of entries for table 1 (second table) excluding the first row
-1     -1     1     1     0     0 ->derivatives
0      1     0     0 ->data
0.2    0.8    0.2    0
0.4    0.6    0.4    0
0.6    0.4    0.6    0
0.8    0.2    0.8    0
1      0     1     0
. . .

```

The term derivatives used here refers to the slope of the krw, kro and pc curve at their start and end. Thus, there are 2 values for each curve and six values in total. A value of zero indicates that the curve

is parallel to the saturation axis. The slope of the $pc(sw)$ curves will always be negative as pc decreases with increasing wetting phase saturation.

Example “rock type.txt” file: 12 model subregions initialized with properties from Middle East Carbonates

Contains 6 linear relative permeability curves for fractures with different apertures, leading to capillary pressures of 100, 200, 1000, and 10000 Pa, respectively:

- fracture 0 - inlet region called WELL (IFT=25 mNm⁻¹, pc =60 Pa)
- fractures (subregions) 1-4 for Kilve model
- fractures (subregions) 6-8 for Arches model

Contains 3 reservoir rock types constituting the subregion integer keys 9-12:

- Intermediate-wet carbonate sample of 4.5 mD permeability and 0.19 porosity, swc =0.21, snr =0.13 (Honarpour, JPT, 1996)
- RT2 oil-wet carbonate sample of 44.9 mD permeability and 0.295 porosity, swc =0.09, sor =0.115 (Gomes et al., SPE ADIPEC, 2008)
- RT3 oil-wet carbonate sample of 24.6 mD permeability and 0.329 porosity, swc =0.07, sor =0.1 (Gomes et al., SPE ADIPEC, 2008)

Table 4: File listing begins here:

```

12
6
-1.00  -1.00  1.00   1.00   0.00   0.00
0.00   1.00   0.00   0.00
0.20   0.80   0.20   0.00
0.40   0.60   0.40   0.00
0.60   0.40   0.60   0.00
0.80   0.20   0.80   0.00
1.00   0.00   1.00   0.00
6
-1.00  -1.00  1.00   1.00   0.00   0.00
0.00   1.00   0.00  100.00
0.20   0.80   0.20  100.00
0.40   0.60   0.40  100.00
0.60   0.40   0.60  100.00
0.80   0.20   0.80  100.00
1.00   0.00   1.00  100.00
6
-1.00  -1.00  1.00   1.00   0.00   0.00
0.00   1.00   0.00  200.00
0.20   0.80   0.20  200.00
0.40   0.60   0.40  200.00
0.60   0.40   0.60  200.00
0.80   0.20   0.80  200.00
1.00   0.00   1.00  200.00

```

```

6
-1.00 -1.00 1.00 1.00 0.00 0.00
0.00 1.00 0.00 10000.00
0.20 0.80 0.20 10000.00
0.40 0.60 0.40 10000.00
0.60 0.40 0.60 10000.00
0.80 0.20 0.80 10000.00
1.00 0.00 1.00 10000.00
6
-1.00 -1.00 1.00 1.00 0.00 0.00
0.00 1.00 0.00 10000.00
0.20 0.80 0.20 10000.00
0.40 0.60 0.40 10000.00
0.60 0.40 0.60 10000.00
0.80 0.20 0.80 10000.00
1.00 0.00 1.00 10000.00
6
-1.00 -1.00 1.00 1.00 0.00 0.00
0.00 1.00 0.00 10.00
0.20 0.80 0.20 10.00
0.40 0.60 0.40 10.00
0.60 0.40 0.60 10.00
0.80 0.20 0.80 10.00
1.00 0.00 1.00 10.00
6
-1.00 -1.00 1.00 1.00 0.00 0.00
0.00 1.00 0.00 13.33
0.20 0.80 0.20 13.33
0.40 0.60 0.40 13.33
0.60 0.40 0.60 13.33
0.80 0.20 0.80 13.33
1.00 0.00 1.00 13.33
6
-1.00 -1.00 1.00 1.00 0.00 0.00
0.00 1.00 0.00 20.00
0.20 0.80 0.20 20.00
0.40 0.60 0.40 20.00
0.60 0.40 0.60 20.00
0.80 0.20 0.80 20.00
1.00 0.00 1.00 20.00
6
-1.00 -1.00 1.00 1.00 0.00 0.00
0.00 1.00 0.00 5.00
0.20 0.80 0.20 5.00
0.40 0.60 0.40 5.00
0.60 0.40 0.60 5.00
0.80 0.20 0.80 5.00
1.00 0.00 1.00 5.00
21
-2.00 0.00 0.00 3.71 -10557.28 -519.57
0.00 1.00 0.00 5000.00
0.05 0.90 0.00 4472.14
0.10 0.80 0.00 3162.28
0.15 0.71 0.00 2581.99
0.20 0.61 0.00 2236.07
0.25 0.53 0.00 2000.00
0.30 0.45 0.01 1825.74
0.35 0.37 0.02 1690.31
0.40 0.30 0.03 1581.14
0.45 0.24 0.04 1490.71
0.50 0.19 0.06 1414.21

```

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0.55	0.14	0.09	1348.40
0.60	0.10	0.13	1290.99
0.65	0.07	0.18	1240.35
0.70	0.05	0.24	1195.23
0.75	0.03	0.32	1154.70
0.80	0.01	0.41	1118.03
0.85	0.01	0.52	1084.65
0.90	0.00	0.66	1054.09
0.95	0.00	0.81	1025.98
1.00	0.00	1.00	1000.00
20			
-2.50	0.00	0.05	0.00 -12339514.50 -2394700.00
0.21	1.000	0.000	275790.29
0.23	0.950	0.001	29000.00
0.24	0.940	0.002	25000.00
0.27	0.860	0.003	15000.00
0.30	0.780	0.009	8000.00
0.33	0.700	0.015	4000.00
0.35	0.640	0.022	3000.00
0.37	0.580	0.028	2000.00
0.40	0.500	0.039	1000.00
0.45	0.380	0.075	-4200.00
0.50	0.250	0.120	-8000.00
0.55	0.160	0.180	-13000.00
0.60	0.090	0.250	-15000.00
0.65	0.050	0.370	-20000.00
0.70	0.030	0.520	-21000.00
0.75	0.015	0.680	-22000.00
0.80	0.006	0.830	-23000.00
0.84	0.002	0.950	-25000.00
0.86	0.000	0.950	-45000.00
0.87	0.000	0.950	-68947.00
30			
0.00	0.00	-41.00	0.00 -7020000.00 -10000000.00
0.09	0.000	0.840	70000.00
0.1	0.000	0.430	-200.00
0.12	0.001	0.300	-2000.00
0.15	0.003	0.180	-2100.00
0.17	0.008	0.120	-2300.00
0.18	0.015	0.100	-2800.00
0.19	0.030	0.090	-2900.00
0.2	0.050	0.080	-3200.00
0.21	0.070	0.070	-4000.00
0.22	0.090	0.058	-4500.00
0.23	0.110	0.054	-5100.00
0.24	0.120	0.052	-5200.00
0.25	0.150	0.045	-5600.00
0.27	0.200	0.030	-5800.00
0.30	0.260	0.020	-7000.00
0.33	0.330	0.015	-8000.00
0.35	0.380	0.010	-10000.00
0.37	0.440	0.008	-12000.00
0.40	0.570	0.006	-14000.00
0.45	0.660	0.005	-16000.00
0.50	0.710	0.003	-20000.00
0.55	0.740	0.001	-24000.00
0.60	0.775	0.000	-32000.00
0.65	0.800	0.000	-38000.00
0.70	0.815	0.000	-49000.00
0.75	0.820	0.000	-65000.00
0.80	0.825	0.000	-92000.00

```

0.85  0.830  0.000  -200000.00
0.88  0.835  0.000  -350000.00
0.885 0.835  0.000  -400000.00
30
0.00  0.00  -11.33 0.00  -2350000.00  -5000000.00
0.07  0.000  0.840  70000.00
0.10  0.000  0.500  -500.00
0.12  0.000  0.350  -6000.00
0.15  0.000  0.220  -10000.00
0.17  0.001  0.150  -10060.00
0.18  0.002  0.130  -10100.00
0.19  0.002  0.100  -11000.00
0.20  0.003  0.090  -11500.00
0.21  0.003  0.070  -12000.00
0.22  0.008  0.059  -12600.00
0.23  0.011  0.054  -12700.00
0.24  0.012  0.050  -13000.00
0.25  0.015  0.045  -15000.00
0.27  0.030  0.030  -15500.00
0.30  0.060  0.020  -16500.00
0.33  0.110  0.015  -17500.00
0.35  0.130  0.012  -18000.00
0.37  0.160  0.011  -19000.00
0.40  0.200  0.010  -19500.00
0.45  0.300  0.009  -22000.00
0.50  0.370  0.007  -25000.00
0.55  0.460  0.004  -30000.00
0.60  0.530  0.003  -36000.00
0.65  0.600  0.002  -45000.00
0.70  0.650  0.001  -52000.00
0.75  0.710  0.000  -62000.00
0.80  0.760  0.000  -80000.00
0.85  0.800  0.000  -150000.00
0.88  0.830  0.000  -250000.00
0.90  0.830  0.000  -350000.00

```

Output from the FECFVM Reservoir Simulator

Apart from the VTK files for visualisation, the simulator also writes a series of text files that are needed for further quantitative analysis of results. The most important file is the “FIP.txt” file. The string FIP here gets appended to the name of the model on which the simulator is run.

The FIP file returns the total volumina of the different model regions and how their total oil and water content changes over time. In the example below, there are just two regions called “matrix” and ‘well’, respectively. These regions desaturate with time as production proceeds.

Table 5: Example FIP output file for a model with the 2 regions named “MATRIX” and “WELL.”

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Time (s)	MATRIX's OIP (m3)	MATRIX's WIP (m3)	WELL's OIP (m3)	WELL's WIP (m3)
0	24.3504	3.0096	0	1.6
5	24.3504	3.00962	0	1.6
10	24.3503	3.00967	0	1.6
20	24.3502	3.00982	0	1.6
30	24.35	3.01001	0	1.6
40	24.3498	3.01022	0	1.6
50	24.3496	3.01044	0	1.6
100	24.3482	3.01177	0	1.6
150	24.3468	3.01322	0	1.6
200	24.3453	3.0147	0	1.6
250	24.3438	3.01623	0	1.6
300	24.3422	3.01779	0	1.6
350	24.3406	3.01941	0	1.6
400	24.3389	3.02106	0	1.6
450	24.3372	3.02277	0	1.6
500	24.3355	3.02452	0	1.6
550	24.3337	3.02631	0	1.6
600	24.3319	3.02815	0	1.6
650	24.33	3.03002	0	1.6
700	24.3281	3.03192	0	1.6
750	24.3261	3.03386	0	1.6
756.641	24.3259	3.03412	0	1.6
750	24.3261	3.03386	0	1.6
800	24.3242	3.03583	0	1.6
850	24.3222	3.03784	0	1.6
900	24.3201	3.03986	0	1.6
950	24.3181	3.04192	0	1.6
1000	24.316	3.04399	0	1.6
1050	24.3139	3.04609	0	1.6